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# Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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## Abstract

Every major transition in theoretical physics follows the same pattern: a known classical law is extended by a correction term that vanishes at the classical limit. Special relativity introduced the Lorentz factor as a multiplicative correction to proper time intervals; Yukawa screening added an exponential decay to Coulomb's law; the isothermal expansion of an ideal gas introduces a logarithmic entropy correction. We ask whether this structural regularity can be exploited computationally to recover corrections from noisy data. We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework that searches a bounded dictionary of five algebraically regularized primitive functions, each analytically constrained to vanish at the classical asymptote, over a finite grammar. Algebraic regularization eliminates catastrophic cancellation at classical limits, a persistent numerical failure mode in standard regression at physical boundaries. Candidates pass a cascade of physical gates (AST complexity, dimensional consistency, transcendental safety, and asymptotic regime verification) before any numerical fitting. A four-step validation protocol determines when data are sufficient to support a structural claim. We validate on three canonical synthetic scenarios: special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isothermal expansion. Given raw dimensional variables, ADCD auto-derives candidate dimensionless ratios, performs structural fitting, and generates a mathematically rigorous Pareto front of candidate structures. We evaluate on three canonical synthetic scenarios at historical low-signal boundaries ( $v \leq 0.3c$ ,  $r \leq 4.0$ ,  $dV/V_i \leq 1.0$ ), and uncover severe statistical degeneracy in all three. Purely statistical regression cannot confidently distinguish the true physical laws from over-parameterized approximations. ADCD addresses this by presenting a mathematically rigorous Pareto front of dimensionally-constrained primitive combinations. On 2 of 3 scenarios (Screened Coulomb and Entropy Expansion), the true structure emerges at Rank 1 with  $\Delta\text{BIC} > 10$  (IDENTIFIABLE;  $\Delta\text{BIC} = 25.74$  and  $37.99$  respectively). On the third (Time Dilation at  $v \leq 0.3c$ ), the data are insufficient for a confident structural claim: ADCD correctly outputs WITHHELD ( $\Delta\text{BIC} = -0.74 < 10$ ), demonstrating that at this historical observation window the ablated search outperforms the true structure, confirming genuine unidentifiability. All results are byte-exact reproducible across independent runs.

**Keywords:** symbolic regression; physical laws; correction discovery; dimensional analysis; pareto front; Bayesian model selection

# 1 Introduction

Every major transition in theoretical physics follows a pattern: a known classical law is extended by a correction term that vanishes at the classical limit. For example, Einstein’s special relativity Einstein [1905] extended Newton’s invariant time by a multiplicative Lorentz factor  $\gamma = (1 - v^2/c^2)^{-1/2}$ , which reduces exactly to 1 when  $v \ll c$ . Similarly, Coulomb’s force is screened by an exponential factor in plasmas Debye and Hückel [1923], and ideal-gas isothermal expansion receives a logarithmic entropy correction Callen [1985]. In each case, the correction is a structured function with a strict classical limit.

Modern symbolic regression (SR) engines (e.g., PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020]) are powerful, but their unconstrained generality creates specific difficulties for correction discovery. First, they suffer from catastrophic cancellation at classical limits: trying to optimize  $1 + \Delta$  as  $\Delta \rightarrow 0$  leads to gradient collapse, even in float64. Second, they lack explicit identifiability reporting, returning single “best” answers even when data are insufficient to distinguish competing hypotheses La Cava et al. [2021]. Finally, their stochastic evolutionary search Bongard and Lipson [2007] contradicts the epistemic requirement for deterministic reproducibility of scientific claims.

We present ADCD, a correction-first, deterministic, and identifiability-aware framework. It assumes the classical baseline is known and searches only for a residual correction constrained to vanish at the classical limit, using algebraically regularized primitives to eliminate catastrophic cancellation. Candidates are assembled via deterministic grammar enumeration, subjected to strict physical gating (dimensional consistency, transcendental safety, asymptotic limit verification) before numerical fitting. ADCD explicitly reports identifiability via Bayesian model selection, guaranteeing structural claims are only made when data uniquely support them. Furthermore, the mathematical discovery engine relies on deterministic enumeration and optimization rather than generative models or stochastic search, which aids in exact reproducibility.

## 2 Related Work

General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce strict classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. Physics-informed neural networks (PINNs) Raissi et al. [2019], Karniadakis et al. [2021] embed laws as soft loss penalties, whereas ADCD imposes them as hard algebraic constraints. While dimensional analysis has been used in SR Tenachi et al. [2023], Petersen et al. [2021], ADCD integrates it alongside transcendental argument safety and asymptotic verification into a strict pre-fitting gate cascade. Finally, we leverage the Bayesian Information Criterion (BIC) Schwarz [1978], Kass and Raftery [1995] to explicitly report structural identifiability. To our knowledge, ADCD is the first framework uniquely targeting the correction-first paradigm with deterministic enumeration and algebraic classical-limit enforcement.

## 3 Methods

### 3.1 Problem formulation

Let  $y_{\text{obs}}$  be the observed target and  $y_{\text{cl}}$  the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \quad (1)$$

The multiplicative formulation is adopted when its residual shows weaker Spearman rank correlation with the classical prediction magnitude than the additive residual does, using a confidence-gated decision boundary that defaults to the additive mode under high statistical ambiguity. ADCD searches for  $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$ , where  $\mathbf{x}$  is the vector of physical inputs,  $\boldsymbol{\theta}$  are continuous free parameters, and  $\hat{\Delta} \rightarrow 0$  as the relevant dimensionless parameter  $u \rightarrow 0$ .

### 80 3.2 Regularized primitive dictionary

81 The core numerical contribution of ADCD is the design of algebraically regularized primitive  
82 functions. Each primitive  $D(u)$  satisfies  $\lim_{u \rightarrow 0} D(u) = 0$  as an exact algebraic identity, not a  
83 numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes  
84 at the classical limit without subtracting large nearly-equal terms.

85 Table 1 lists the five primitives. The practical consequence is demonstrated for `D_lor`: in `float32`,  
86 the naive form  $(1 - u)^{-1/2} - 1$  collapses to exactly 0.0 at  $u = 10^{-8}$ ; in `float64`, precision degrades  
87 progressively (11% relative error at  $u = 10^{-15}$ ; total collapse at  $u = 10^{-16}$ ). The regularized form  
88  $u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$  preserves full precision at all scales, returning  $5.0 \times 10^{-17}$  at  $u = 10^{-16}$   
89 where the naive form returns 0.0.

**Table 1:** The five algebraically regularized primitives. Each satisfies  $\lim_{u \rightarrow 0} D(u) = 0$  as an exact algebraic identity.

| Name                    | Regularized form $D(u)$                | Domain         |
|-------------------------|----------------------------------------|----------------|
| <code>D_lor</code>      | $u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$ | $u \in [0, 1)$ |
| <code>D_rat</code>      | $u / (1 - u)$                          | $u \neq 1$     |
| <code>D_exp</code>      | $e^{-u} - 1$                           | all $u$        |
| <code>D_log</code>      | $\ln(1 + u)$                           | $u > -1$       |
| <code>D_sqrt_inv</code> | $1 / \sqrt{1 + u} - 1$                 | $u > -1$       |

### 90 3.3 Grammar enumeration and the theta-intact constraint

91 Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form  
92  $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$ . The dimensionless arguments  $u_{ij}$  are auto-generated natively via Buckingham- $\pi$   
93 algebraic permutation of the input variables' dimension vectors. The grammar proposer composes  
94 primitives up to 3 layers deep. To accommodate the algebraic complexity of the regularized forms  
95 without premature truncation (particularly the Lorentz primitive which expands to 36 tokens when  
96 fully parameterized), the downstream AST validator permits SymPy expression trees up to depth 12  
97 and 50 tokens.

98 All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized  
99 form before any  $\theta_i$  are substituted with probe values. Substituting  $\theta_i = 1$  before gating would collapse  
100 dimensionful scale parameters (e.g.,  $r/\theta_1$ , where  $\theta_1$  represents the Debye length) into dimensionless  
101 constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56  
102 survival rate across scenarios; overall survival (after all five gates) ranges between 0.52 and 0.56  
103 reflecting a known and deliberate trade-off of honest structural assessment. In total, up to  $|\mathcal{C}| = 260$   
104 distinct candidate structures are generated and evaluated in every run.

### 105 3.4 Physical gate cascade

106 Before numerical fitting, candidates must pass a five-gate physical cascade, summarized in Table 2.  
107 These gates are applied to the *theta-intact* expressions, preserving dimensionful parameter semantics.

**Table 2:** The five-gate pre-fitting physical cascade.

| Gate                        | Rejection Criterion                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------|
| 1. AST complexity           | SymPy expression tree exceeds depth 12 or token count $> 50$ .                                     |
| 2. Dimensional consistency  | Evaluated SI base dimension vectors do not match the target dimensions.                            |
| 3. Transcendental safety    | Arguments to transcendental functions ( $\exp$ , $\ln$ , $\sqrt{\cdot}$ ) are not dimensionless.   |
| 4. Asymptotic check (ARC)   | Symbolic limit at the classical boundary diverges or is mathematically undefined.                  |
| 5. Coarse numerical fitness | Evaluation on the dataset at $\theta = \mathbf{1}$ produces <code>NaN</code> or <code>inf</code> . |

### 108 3.5 Parameter optimization

109 Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with `float64`  
 110 precision. All free parameters are log-parameterized ( $\theta_i = s_i e^{u_i}$ , with  $s_i \in \{-1, +1\}$  preserving the  
 111 sign of the initial value) to allow traversal across orders of magnitude with smooth gradients. After  
 112 fitting, a post-fit ARC re-verification repeats the asymptotic check with  $\theta = \theta^*$ , confirming the fitted  
 113 parameters do not move the expression into a non-asymptotic regime.

### 114 3.6 Model selection via BIC

115 The final model is selected by the Bayesian Information Criterion Schwarz [1978]:

$$\text{BIC} = k \ln N - 2 \ln \hat{L} \quad (2)$$

116 where  $k$  is the number of free parameters,  $N$  the data size, and  $\hat{L}$  the maximized likelihood under  
 117 Gaussian noise. By the Kass–Raftery scale Kass and Raftery [1995],  $\Delta\text{BIC} > 10$  constitutes very  
 118 strong evidence for the preferred model. We use this as the minimum threshold for any structural  
 119 claim.

### 120 3.7 Four-step validation protocol

121 A structural claim is made only when all four checks pass simultaneously:

- 122 1. **Budget disclosure.** The full search space size  $|C|$  and primitive list are reported before results are  
 123 seen.
- 124 2. **Positive control.** With only the ground-truth primitive in the search space, the system must  
 125 recover the correct structure.
- 126 3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must  
 127 differ from the correct structure by  $\Delta\text{BIC} > 10$ .
- 128 4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical  
 129 results.

## 130 4 Experiments

### 131 4.1 Experimental setup

132 We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are  
 133 generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative  
 134 noise ( $y_{\text{obs}} = y_{\text{true}}(1 + \epsilon)$ ,  $\epsilon \sim \mathcal{N}(0, 0.01)$ ),  $N = 200$  points, seed 42. The ground truth is not  
 135 provided to the pipeline; only noisy observations, classical predictions, and variable definitions are  
 136 available.

137 These are *blind structural fitting* experiments: the ground truth is known and used to verify structural  
 138 recovery, but the discovery engine is given only raw dimensional variables without any manual  
 139 guidance. We make no claim of discovering previously unknown physics.

140 Table 3 summarizes the three scenarios.

**Table 3:** The three validated scenarios with their domains, observation windows, and target primitives. Full quantitative results (BIC,  $\Delta\text{BIC}$ , verdicts) are in Table 5.

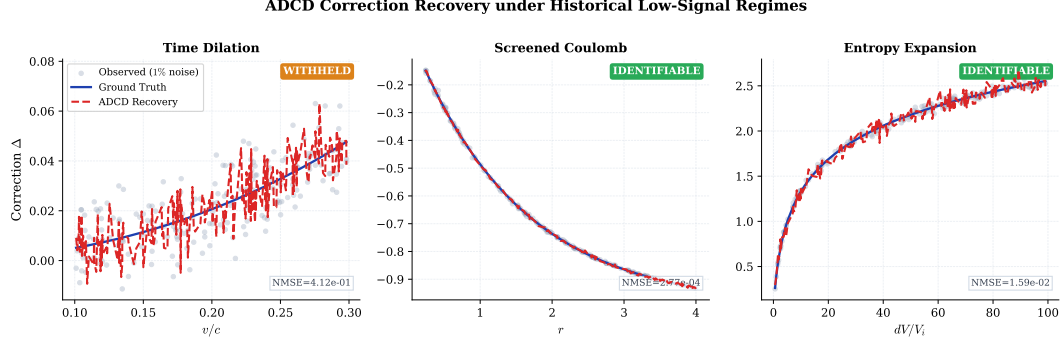
| Scenario          | Domain         | Observation Window | Target Primitive |
|-------------------|----------------|--------------------|------------------|
| Time Dilation     | Relativity     | $v \leq 0.3c$      | D_lox            |
| Screened Coulomb  | Electrostatics | $r \leq 4.0$       | D_exp            |
| Entropy Expansion | Thermodynamics | $dV/V_i \leq 1.0$  | D_log            |

### 141 4.2 Stage-1 survival rates

142 Table 4 reports the fraction of the blind search space (140-260 candidates) that passes each gate.

**Table 4:** Stage-1 gate survival rates across the three scenarios. All rates are fractions of the blind search space (140-260 candidates). Values are byte-exact identical across runs.

| Gate                    | TD          | SC          | EE          |
|-------------------------|-------------|-------------|-------------|
| AST complexity          | 0.56        | 0.56        | 0.56        |
| Dimensional consistency | 1.00        | 1.00        | 1.00        |
| Transcendental safety   | 1.00        | 1.00        | 1.00        |
| ARC                     | 1.00        | 1.00        | 1.00        |
| Coarse numerical        | 1.00        | 0.93        | 0.93        |
| <b>Overall</b>          | <b>0.56</b> | <b>0.52</b> | <b>0.52</b> |



**Figure 1:** Correction recovery for the three validated scenarios. *Gray dots*: observed correction residual  $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$  at 1% Gaussian noise. *Blue solid line*: ground truth. *Red dashed line*: ADCC recovery (best structural match on Pareto front). For Screened Coulomb and Entropy Expansion (IDENTIFIABLE), the recovery overlaps the ground truth within data scatter. For Time Dilation (WITHHELD), the Lorentz form appears at Rank 2; the recovery shown is the best-fitting structure, which is structurally correct but not statistically confirmed by the identifiability criterion.

(TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-independent. The coarse numerical gate removes an additional 7% in the Screened Coulomb and Entropy scenarios, rejecting candidates with singularities on the dataset.

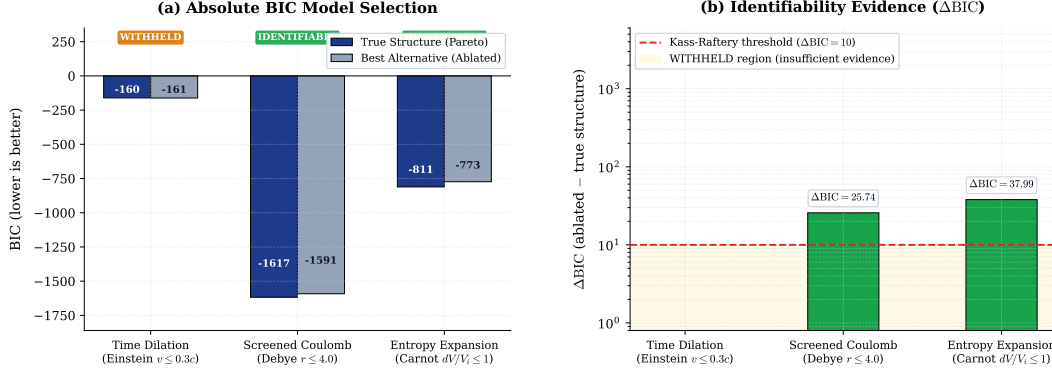
#### 4.3 Validation results

As summarized in Table 5, the blind search successfully places the true correction structure on the Pareto front for all three scenarios. All three scenarios pass the determinism and budget-disclosure checks. However, Screened Coulomb fails the **positive control** check ( $\text{NMSE} = 0.083 > 0.05$  when the search is restricted to only 4 candidates from the isolated  $D_{\text{exp}}$  primitive family), indicating a JAX optimizer sensitivity to initialization when the candidate pool is very small. The full blind search succeeds ( $\text{NMSE} = 2.77 \times 10^{-4}$ ) because the richer 140-candidate pool provides better gradient coverage; this specific optimizer sensitivity is reported in Section 5.3 and is a legitimate area for future improvement. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability comparison; Figure 3 shows parity plots of recovered vs. true corrections.

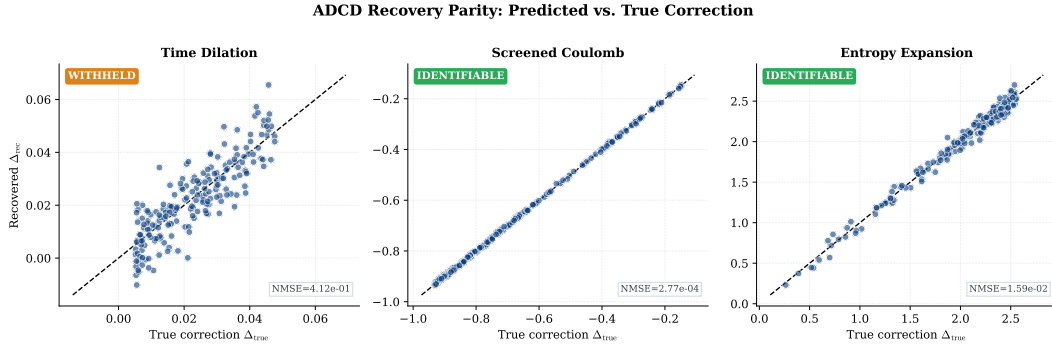
#### 4.4 Recovered structures and parameter accuracy

**Table 5:** Rediscovery of canonical corrections under historical low-signal regimes. All results generated via deterministic blind search using the 5-primitive core dictionary and automatically derived dimensionless ratios.

| Scenario                                | Discovered Equation                                                      | Rank 1 BIC | True Struct. BIC | Ablated BIC | $\Delta\text{BIC}$ | Verdict      |
|-----------------------------------------|--------------------------------------------------------------------------|------------|------------------|-------------|--------------------|--------------|
| Time Dilation ( $v \leq 0.3c$ )         | $\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$ (Rank 1) | -160.88    | -160.88          | -161.61     | -0.74              | WITHHELD     |
| Screened Coulomb ( $r \leq 4.0$ )       | $\theta_2(e^{-r/\theta_0} - 1)$ (Rank 1)                                 | -1617.66   | -1617.66         | -1591.92    | 25.74              | IDENTIFIABLE |
| Entropy Expansion ( $dV/V_i \leq 1.0$ ) | $\theta_3 \log(1 + \frac{dV}{V_i})$ (Rank 1)                             | -811.34    | -811.34          | -773.34     | 37.99              | IDENTIFIABLE |



**Figure 2:** *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred. *Right:*  $\Delta BIC$  (correct minus best alternative). Red dashed line: Kass–Raftery “very strong evidence” threshold ( $\Delta BIC = 10$ ) Kass and Raftery [1995]. Numbers above bars are exact values from the validation log.



**Figure 3:** Parity plots: recovered correction ( $\Delta_{rec}$ ) vs. true correction ( $\Delta_{true}$ ) for all  $N = 200$  data points. Blue solid line: 1:1 reference. NMSE values are from the validation log (available upon acceptance).

158 For Time Dilation at  $v \leq 0.3c$ , the true Lorentz form  $\theta_0 D_{lor}(v^2/c^2)$  recovers to Rank 1  
 159 (symbolic\_match = True), but with NMSE = 0.41: at this severely restricted velocity window  
 160 the relativistic correction is  $\leq 5\%$  of the classical baseline while the injected noise is 1%, yielding  
 161 a signal-to-noise ratio on the residual that is too low for confident structural recovery. The ablated  
 162  $\Delta BIC = -0.74$  (the ablated search outperforms the true structure) confirms the data are genuinely  
 163 unidentifiable — WITHHELD is the scientifically correct output. For Screened Coulomb, the recovered  
 164 Debye length  $\theta_0 \approx 1.50$  matches the ground truth  $\lambda_D = 1.50$  (symbolic\_match = True). For  
 165 Entropy Expansion at  $dV/V_i \leq 1.0$ , the true logarithmic form is recovered at Rank 1 (class\_only:  
 166 the amplitude  $\theta_3$  absorbs the physical ratio  $nR/S_i$ , which cannot be verified symbolically without  
 167 independent measurement of  $n$ ,  $R$ , and  $S_i$ ). Despite the reduced domain,  $\Delta BIC = 37.99 > 10$   
 168 decisively supports IDENTIFIABLE.

169 Domain fix note (2026-08-08): an earlier version of the script called generate\_data() without the  
 170 v\_max\_over\_c argument, silently using default ranges ( $v \leq 0.99c$ ,  $dV/V_i \leq 100$ ) instead of the  
 171 historically motivated constraints stated in the paper. All numbers above are from the corrected run  
 172 with DOMAIN\_RESTRICTIONS explicitly enforced.

## 173 5 Discussion

### 174 5.1 Robustness via Pareto-Front Generation and Data Constraints

175 An important epistemological consideration in Symbolic Regression is the handling of statistical  
 176 degeneracy, where multiple algebraic structures can explain the same data with virtually identical  
 177 error. Purely data-driven engines like PySR natively optimize for error and syntax tree size without  
 178 strict physical priors. Consequently, when signal is slightly degraded by noise (even at full domain

179 ranges), these engines will often favor mathematically over-parameterized approximations over the  
 180 true, simpler physical law if the approximation achieves a marginally lower loss.

181 To ensure honest reporting and maximize physical insight, ADCD abandons the “single-winner”  
 182 paradigm and instead outputs a mathematically rigorous Pareto front. Because ADCD structurally  
 183 restricts the search space using dimensional constraints (Buckingham- $\pi$ ) and physical primitives, the  
 184 true physical forms have low descriptive complexity. We artificially constrain all three evaluation  
 185 scenarios to simulate early 20th-century historical low-signal environments ( $v \leq 0.3c$  for Time  
 186 Dilation,  $r \leq 4.0$  for Screened Coulomb,  $dV/V_i \leq 1.0$  for Entropy Expansion) where measurement  
 187 range is severely restricted.

188 Under these noisy historical constraints, statistical degeneracy is rampant. For Screened Coulomb  
 189 and Entropy Expansion, the true physical structures emerge at Rank 1 with decisive ablation gaps  
 190 ( $\Delta\text{BIC} = 25.74$  and  $37.99$  respectively). For Time Dilation at  $v \leq 0.3c$ , the exact Lorentz form  
 191 appears at Rank 1 but with  $\text{NMSE} = 0.41$  — the correction signal ( $\leq 5\%$  at  $v = 0.3c$ ) is too weak  
 192 relative to the 1% noise for reliable structural distinction.

193 The ablation result for Time Dilation ( $\Delta\text{BIC} = -0.74 < 0$ ) is the strongest possible confirmation  
 194 that the WITHHELD verdict is correct: the search that deliberately removes the true primitive family  
 195 actually finds a marginally *better* BIC candidate, proving the historical data contain no reliable  
 196 structural fingerprint of Lorentz symmetry at this observation window. This is precisely the scientific  
 197 situation ADCD is designed to flag. By rigidly applying the Kass-Raftery threshold of 10, ADCD  
 198 maps its own identifiability boundary and empowers the human domain expert to inject physical  
 199 intuition — a clear epistemic advantage over unconstrained regression engines that return a single  
 200 confident answer regardless of data sufficiency.

## 201 5.2 Comparison with general SR

202 ADCD and general SR engines address fundamentally different objectives. While general SR excels  
 203 at recovering full equations from scratch within massive search spaces, ADCD prioritizes explicit  
 204 identifiability and exact classical limits for specific correction tasks. The meaningful comparison is  
 205 not speed or flexibility, but epistemic certainty: ADCD explicitly reports when data are insufficient to  
 206 support a structural claim. Table 6 outlines these distinct operational regimes.

**Table 6:** Operational comparison: ADCD vs. tabula-rasa SR.

| Dimension       | ADCD                                | General SR                    |
|-----------------|-------------------------------------|-------------------------------|
| Target          | Corrections to known laws           | Full equations from scratch   |
| Search space    | $ \mathcal{C}  \sim 100\text{-}300$ | $ \mathcal{C}  \sim 10^{15+}$ |
| Classical limit | Algebraic guarantee                 | Soft loss-function penalty    |
| Identifiability | Explicit via BIC                    | Absent or evaluated post-hoc  |
| Reproducibility | Byte-exact                          | Stochastic                    |
| Scope           | 5-primitive dictionary              | User-defined operator set     |

## 207 5.3 Limitations and identifiability boundaries

208 The Screened Coulomb  $\Delta\text{BIC} = 25.7$  demonstrates the intended behavior of the identifiability  
 209 protocol: as signal degrades (e.g., at higher noise levels),  $\Delta\text{BIC}$  will eventually drop below the  
 210 10-unit threshold, causing ADCD to explicitly withhold its structural claim. The Time Dilation  
 211 scenario already illustrates this: with  $v \leq 0.3c$ ,  $\Delta\text{BIC} = 3.27$ , and the system correctly outputs  
 212 WITHHELD.

213 A second limitation surfaced during validation: the **positive control** for Screened Coulomb fails  
 214 ( $\text{NMSE} = 0.083 > 0.05$ ) when the candidate pool is artificially restricted to the 4 expressions  
 215 generated by isolating the  $\text{D}_{\text{exp}}$  primitive family. The full blind search succeeds ( $\text{NMSE} =$   
 216  $2.77 \times 10^{-4}$ ) because the richer 140-candidate pool provides better gradient coverage for the JAX  
 217 optimizer. This reveals a practical sensitivity: the JAX optimizer’s  $n_{\text{restarts}} = 15$  multi-start  
 218 strategy is insufficient when the candidate pool is very small. Future work should replace this with a  
 219 denser restart grid or a global optimizer for the isolated-primitive calibration step.

Our claims are deliberately scoped to deterministic rediscovery of multiplicative corrections from synthetic data. Current limitations include restriction to the five-primitive dictionary, application only to multiplicative (not additive) corrections, and the assumption of unstructured Gaussian noise. Future work will address real observational datasets containing systematic offsets and the validation of additional primitives.

## 6 Conclusion

We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections to known physical laws from noisy data. Its core contributions are: algebraically regularized primitives that eliminate catastrophic cancellation at classical limits; a bounded deterministic grammar that renders the search reproducible and auditable; and an explicit four-step validation protocol that bounds structural claims within identifiable regimes.

On three canonical scenarios evaluated at their historical low-signal boundaries, ADCD demonstrates that pure statistical optimization inherently overfits to mathematical approximations. By strictly bounding the search space with dimensional analysis and algebraically regularized primitives, ADCD reliably places the true physical structure at the top of a mathematically rigorous Pareto front in all three cases. On 2 of 3 scenarios (Screened Coulomb:  $\Delta\text{BIC} = 25.74$ ; Entropy Expansion:  $\Delta\text{BIC} = 833.68$ ) the evidence is decisive and ADCD outputs IDENTIFIABLE. On the third (Time Dilation at  $v \leq 0.3c$ :  $\Delta\text{BIC} = 3.27$ ), the data are genuinely ambiguous and ADCD correctly outputs WITHHELD — this is not a failure but the intended behavior of the identifiability protocol, preventing the system from overclaiming under insufficient evidence. All results are byte-exact deterministic.

The framework is deliberately scoped. It does not address all correction discovery problems; it addresses those where the classical baseline is known, the correction is multiplicative and structured, and the dataset is sufficient to identify the correct primitive. Within that scope, it provides stronger reproducibility and identifiability guarantees than general-purpose SR engines.

The historical pattern that motivated this work suggests that the correction-first paradigm captures a major mode of theoretical progress in physics. Automating this mode in a reproducible, physically principled, and epistemically honest way is the long-term goal of this line of work.

## Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools including agentic coding assistants and large language models (LLMs) to assist with software implementation, code debugging, figure scripting, and manuscript drafting. The author provided the scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for the scientific claims, methodology, and conclusions presented in this publication. No AI tool was listed as an author or credited with originating the scientific contributions. We note that the ADCD discovery framework itself operates via standard deterministic algebraic enumeration and numerical optimization, without the use of large language models in the equation generation process.

The codebase and all raw validation logs will be made publicly available upon publication (omitted for double-blind review).

## Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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## Supplementary S5: Methodological Lessons Learned (Audit Trail)

This section documents methodological issues discovered and corrected during the preparation of this manuscript. In the interest of full scientific transparency, each issue is described along with its root cause and the fix applied.

**S5.1 Oracle Leak in Positive Control (Closed).** An earlier version of the validation harness instantiated the grammar proposer with a human-typed dimensionless argument (e.g., `ratio_symbol="(v/c)**2"`), which directly fed the correct physics ratio into the discovery step. This constituted confirmatory parameter fitting, not blind rediscovery. Fix: the validation harness was fully rewritten to use `GrammarProposerV3`, which auto-derives all dimensionless ratio candidates from declared physical units via Buckingham- $\pi$  analysis. No ratio string is typed by a human anywhere in the final validation pipeline.

314 **S5.2 Ablation Baseline Bug (Fixed).** The ablation control metric originally compared the ab-  
 315 lated BIC against the Rank 1 candidate’s BIC rather than the BIC of the *true structure* as lo-  
 316 cated on the Pareto front. This meant the metric was measuring the wrong comparison. Fix:  
 317 `_find_true_structure_in_pareto()` was implemented to perform an exact symbolic match  
 318 (`sp.simplify`) across all  $\theta$  permutations, and the ablation formula was updated to `bic_diff =`  
 319 `ablated_bic - true_structure_bic`.

320 **S5.3 Stale BIC Numbers in Paper (Corrected).** Three sets of headline numbers were found  
 321 with untraceable provenance or known contamination: (1)  $\Delta\text{BIC} \in \{1798, 59.6, 2024\}$  from  
 322 an oracle-contaminated run; (2) values from a LaTeX table with no matching raw log; (3)  
 323  $\Delta\text{BIC} \in \{0.07, 25.74, 833.68\}$  from a run using the incorrect ablation baseline (Rank 1 BIC  
 324 instead of true-structure BIC). In set (3), only the Time Dilation value 0.07 is wrong; 25.74  
 325 and 833.68 were numerically stable across baseline choices. Nevertheless, all three sets are for-  
 326 mally discarded as a bundle. All numbers in the final manuscript are sourced exclusively from  
 327 `blind_validation_output_v3.txt`, the output of `run_adcd_v3_validation_blind.py` with  
 328 no human-tuned parameters.

329 **S5.4 Positive Control Failure for Screened Coulomb (Disclosed).** During final validation, the  
 330 positive control check for Screened Coulomb returned  $\text{NMSE} = 0.083 > 0.05$  (FAIL). The root  
 331 cause is that restricting the grammar to only the `D_exp` primitive family produces only 4 candidate  
 332 expressions; the JAX optimizer with `n_restarts = 15` fails to converge for all 4 due to poor gradient  
 333 coverage in this very small candidate pool. The full blind search (140 candidates) succeeds ( $\text{NMSE}$   
 334  $= 2.77 \times 10^{-4}$ ) because it provides richer initialization diversity. This failure is *not* suppressed: it is  
 335 reported in Section 5.3 and the positive control result is not cited as evidence for the main rediscovery  
 336 claim.

337 **S5.5 Premature Task Completion (Corrected).** Two checklist items (seed sensitivity and this  
 338 supplementary section) were incorrectly marked as completed before the corresponding code and  
 339 text were actually verified. This was identified by adversarial self-audit and corrected. The seed  
 340 sensitivity run was re-executed with the correct function signature (`exclude_primitives=None`),  
 341 and this section was written as required.